Landfill methane recovery and use under deep uncertainty: a satellite-constrained Monte Carlo decision model for the Norte III landfill, Buenos Aires

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Highlights

- Satellite-constrained methane balance of a megasite, calibrated in two periods
- Collection efficiency is inferred at about 14%, not measured; 3 error models agree
- No integrated intervention has a positive median NPV without carbon credits
- Added engines on surplus gas pay; biomethane needs about US$14 per MMBtu
- Cover helps only if captured share of blocked gas exceeds an analytic threshold

Abstract

Very large landfills are major methane sources, but site decisions rarely combine operator-reported capture, independent emission estimates and the economics of competing gas uses. We built a Latin-hypercube Monte Carlo decision model (300,000 draws, 78 inputs) of landfill-gas generation, collection, surface emission and oxidation, selective covers, biomethane upgrading, additional power generation and photovoltaics, and applied it to the Norte III landfill in Buenos Aires. A bottom-up generation prior is constrained by satellite emission estimates at their own observation date (2022, with the capture of that period) and carried forward to the 2025–26 decision period; three observation-error models are compared. The inferred collection efficiency is 0.13 (0.09 to 0.19) in 2022 and 0.14 (0.10 to 0.21) in 2025–26 (median, 90% interval), a model output conditional on the priors, not a measurement. Methane, carbon and energy balances close for every scenario. Under the assumed no-credit conditions, none of the eight integrated intervention scenarios has a positive median incremental net present value; wellfield optimisation with a selective cover abates methane at a median US$3.6 per tCO_2e. Additional engines on surplus gas, reported separately, have positive value in 99% of draws at merchant tariffs, whereas biomethane needs a bio-GNC plant-gate price of US$0.46 per Nm^3 (about US$14 per MMBtu) to break even. A cover reduces net emissions only when the captured share of blocked gas exceeds an analytic break-even (median 0.14). Conclusions are unchanged under correlated inputs, alternative priors, time-varying generation and dated Argentine price, tax and financing benchmarks.

Keywords

Landfill gas; Methane mitigation; Satellite constraint; Biomethane; Monte Carlo decision analysis; Techno-economic analysis; Argentina

1. Introduction

Landfills are among the largest anthropogenic sources of methane, and airborne and satellite surveys have shown that a small number of very large sites contribute a disproportionate share of urban emissions (Duren et al., 2019; Maasakkers et al., 2022; Schuit et al., 2023; Cusworth et al., 2024). Inventories derived from first-order decay models and assumed collection efficiencies can differ substantially from measurement-based estimates (Stuart et al., 2026), and collection efficiency itself is poorly constrained in the literature (Spokas et al., 2006; Giordano et al., 2024; Saul et al., 2026). The Norte III landfill in Greater Buenos Aires is one of the large sites identified from space. Its operator reports a large and growing capture system with on-site power generation (CEAMSE, 2026; Canal 26, 2026), while satellite observations place surface emissions at tens of tonnes of methane per hour (Maasakkers et al., 2022; Bloomberg News, 2022). Both statements can hold only if most of the methane generated is not collected, and a decision model for the site must respect that tension.

Mitigation options are numerous: wellfield tuning and leak repair, engineered covers, new uses of recovered gas (additional engines, compressed bio-GNC for vehicles, pipeline biomethane), on-site solar generation and gas buffering. Techno-economic studies usually evaluate such options one at a time, with point estimates (Ghafoori et al., 2022; Agostini et al., 2024). Three gaps follow. First, the climate benefit of a measure is rarely tied to a mass balance in which blocked or captured gas can reappear elsewhere. Second, uncertainty in prices, exchange rates, taxes and financing is usually represented by a few scenarios rather than propagated, and site-relevant prices are seldom dated. Third, the site's emission estimate is rarely confronted with independent top-down data, so the climate value of added capture is uncertain in a way that is not reported.

This paper contributes a satellite-constrained decision model for a single landfill. To our knowledge, no earlier study of a landfill combines (i) a site methane balance constrained by satellite emission estimates at their own observation date, with explicit alternative observation-error models, (ii) a stochastic comparison of the competing uses of recovered gas in which every scenario closes methane, carbon and energy balances, and (iii) closed-form break-evens for selective covers (the minimum captured share of blocked gas) and for photovoltaic coatings (the minimum efficiency), verified against the simulation. Economic inputs are anchored to dated Argentine benchmarks, and the results are tested against correlated inputs, alternative priors, structural alternatives and time-varying generation.

The study is a screening analysis. It uses no site measurements beyond published operator and satellite figures, it treats prices and unit costs as distributions conditioned on dated benchmarks, and it makes no claim about what the operator will do. Its purpose is to show which decisions are robust to the stated uncertainty, which are not, and which measurements would resolve them.

2. Case study, reported data and benchmarks

2.1 Operator data and top-down estimates

Norte III receives roughly 18,000 t of municipal solid waste per day (CEAMSE, 2026). CEAMSE reports captured biogas of 13,800 $m^3\ h^{-1}$ in 2025, rising to 16,000 $m^3\ h^{-1}$ in April 2026 (Canal 26, 2026), against a maximum captured flow of 13,100 $Nm^3\ h^{-1}$ in 2013 (Rosso, 2016). Biogas-fired power plants have operated on site since 2012 (Canal 26, 2026); the operator statement summarised in the project white paper (Sandler, 2026) gives 23 MW installed in 2020 and an expansion to 33 MW planned for 2026. Gas not used by the engines is flared. These are operator

statements: they are not measured fugitive emissions, they do not state the methane fraction, and they do not establish that any gas is uncommitted.

Top-down estimates come from satellites. Maasakkers et al. (2022) combined TROPOMI and GHGSat observations and reported emissions of 3 to 29 t CH_4 h^{-1} from strongly emitting landfills in four cities, with the Buenos Aires site near 28 t h^{-1} (SRON, 2022); a GHGSat observation in September 2022 implied 18 to 38 t h^{-1} (Bloomberg News, 2022). These estimates refer to 2019–2022. Capture has since increased, and comparing them with a 2025–26 capture rate, as the first version of this analysis did, mixes periods. CEAMSE disputes the characterisation of the site (Canal 26, 2026). The satellite estimates are therefore used as a constraint at their own date, with an explicit observation-error model, not as ground truth (Section 3.2).

2.2 Dated Argentine economic benchmarks

Table 2 lists the market and regulatory benchmarks to which the economic priors are anchored, with their dates. Gas values distinguish the wellhead (PIST) price that a pipeline injector would receive, the delivered cost of gas to compressed natural gas (GNC) stations in the Buenos Aires metropolitan area and the retail GNC price; a bio-GNC plant supplying an anchor fleet would sell between the second and the third, net of compression and station costs. Electricity values distinguish the merchant (wholesale) value of energy from the long-term contract tariffs that the RenovAr programme awarded to landfill-gas projects. Peso values are converted at an assumed 1,400 ± 100 ARS per USD (late 2025) and should be re-dated before use. Capital goods carry import duties that Decree 513/2025 reduced to 12.6% for many items, against the 20% Mercosur common external tariff (Marval O'Farrell Mairal, 2025); the imported share, duty and logistics define the capital premium, Eq. (9). Country risk enters through the real discount rate, and corporate income tax, the debt share and credit eligibility are treated in the robustness analysis (Section 4.8).

Item	Benchmark	Date	Source	Prior used
Wellhead (PIST) gas price, AMBA distributors	US$2.94–2.98 per MMBtu	Dec 2025	(Estudio O'Farrell, 2025)	p_pipe tri(0.07, 0.10, 0.16) US$ per Nm^3
Gas delivered to AMBA GNC stations (pre-VAT)	ARS 221.63 per m^3 (≈ US$0.16)	Nov 2025	(Surtidores, 2025)	p_gnc tri(0.13, 0.19, 0.40) US$ per Nm^3
Retail GNC price, AMBA	ARS 500–570 per m^3 (≈ US$0.36–0.41)	Jul 2025	(La Nación, 2025)	upper end of p_gnc
Wholesale electricity: marginal and priority-market cost	US$69 and 64 per MWh	2024	(MEGSA, 2025)	p_el tri(60, 75, 95) US$ per MWh (merchant)
Monomic wholesale electricity price	ARS 121,100 per MWh (≈ US$87)	Aug 2025	(UDEA, 2025)	c_el tri(70, 95, 130) (purchase incl. network)
RenovAr landfill-gas contract awards	US$128–130 per MWh	2017	(Energías Renovables, 2017; Renewable Energy Magazine, 2017)	p_el,k tri(110, 125, 135) (contract variant)
Import duty on capital goods	12.6% (Decree 513/2025); 20% CET	Jul 2025	(Marval O'Farrell Mairal, 2025)	duty tri(0, 0.126, 0.20)
Country risk (EMBI spread)	561 to >1,200 bp; 569 bp on 17 Dec	2025	(USAL, 2025)	r tri(0.08, 0.12, 0.18) real
Corporate income tax, top bracket	35%	2021 law	(Argentina, 2021)	after-tax variant
Landfill-gas carbon credits; voluntary-market average	≈US$9.5 and 5.5 per tCO_2e	2025	(MSCI, 2025)	cp tri(2, 6, 12), eligibility uni(0.2, 0.8)

Table 2. Dated Argentine benchmarks and the priors anchored to them (US$ conversions at 1,400 ARS per USD, an assumption).

3. Methods

3.1 Methane balance in two periods

Gas quantities are in Nm³ CH_4 h^{-1} unless stated. Generation around the observation date, G_{22}, is bottom-up, from waste acceptance A (t d^{-1}), methane potential L_0 (Nm³ CH_4 t^{-1}) and an effective fraction f_{ss} of steady-state generation that absorbs landfill age, decay rate and lag; the prior is deliberately wide (L_0 from 50 to 170, f_{ss} from 0.30 to 0.95; Appendix A):

$$G_{22} = \frac{365\ \ A\ \ L_0\ \ f_{ss}}{8760}, \qquad G = g_t\ \ G_{22} \tag{1}$$

where g_t (0.95 to 1.15) carries generation forward to the 2025–26 decision period. Collected methane is the reported biogas flow times the methane fraction, separately for the historical period (Q_{22}, 12,000 to 14,000 Nm³ h^{-1}, between the 2013 peak and the 2025 value) and the decision period (Q, 13,000 to 16,500 Nm³ h^{-1}). The collection efficiency of each period is an inferred quantity, $\eta = C/G$, and draws above 0.95 are rejected as inconsistent. Surface emission after oxidation in the cover soil, a fraction ox (Chanton et al., 2009), is

$$E = \left(G - C\right)\left(1 - ox\right) \tag{2}$$

evaluated with G_{22} and C_{22} at the observation date and with G and C for the decision period. Existing engines consume $P_{el}\,CF/(\eta_{el}\,LHV)$, with LHV = 9.94 kWh Nm^{-3}; the surplus beyond the engines, S, is flared. Methane is converted with 0.717 kg Nm^{-3} and to CO_2-equivalent with the AR6 values for non-fossil methane, 27.0 (GWP100) and 79.7 (GWP20) (Forster et al., 2021; Balcombe et al., 2018).

3.2 Satellite constraint and observation-error models

The satellite information enters as a likelihood on the modelled emission at the observation date, E_{22} (t h^{-1}). Three observation-error models are compared: a uniform window over the reported range (18 to 38 t h^{-1}; I is the indicator function), a Gaussian centred on the peer-reviewed estimate (28 ± 6 t h^{-1}), and a lognormal with the same median and $\sigma_{\ln}$ = 0.35, the base case because emission-rate retrievals have multiplicative errors (Schuit et al., 2023):

$$L_{\text{win}} = \mathrm{I}\left(18 \le E_{22} \le 38\right), \quad L_{\text{G}} = e^{-\frac{(E_{22}-28)^2}{2\cdot 6^2}}, \quad L_{\text{LN}} = e^{-\frac{(\ln E_{22}-\ln 28)^2}{2\cdot 0.35^2}} \tag{3}$$

Posterior samples are drawn by importance resampling of the prior draws with weights $w_i \propto L(E_{22,i})$ (Beaumont et al., 2002); the effective sample size is

$$\mathrm{ESS} = \frac{\left(\sum_i w_i\right)^2}{\sum_i w_i^2} \tag{4}$$

which is 224,817 (lognormal), 186,963 (Gaussian) and 182,249 (window) of 300,000 draws. The window model is equivalent to approximate Bayesian rejection. Every collection efficiency reported below is an inference conditional on the generation prior and the error model; none is a measurement.

3.3 Interventions, selective covers and the cover break-even

Eight scenarios follow the structure of the project white paper (Table 1). Wellfield optimisation (S1) raises the collection efficiency by an absolute increment $\Delta\eta_1$; the added gas is flared unless a later scenario uses it. A selective cover (S2, S3) acts on the fraction fh_{ot} of surface flux passing through the covered area. With M_c the methane arriving at the cover, the barrier blocks $b = \varepsilon M_c$; a share s_c of the blocked gas is captured by the extraction system and the rest is redirected to uncovered surface with oxidation $\rho\ ox$, while oxidation under the cover falls to $ox_{cov} = ox(1-\delta\varepsilon)$. The emission after the cover and the net methane avoided are

$$E_1 = (E_s - M_c)(1 - ox) + (M_c - b)(1 - ox_{cov}) + (1 - s_c)\ b\ (1 - \rho\ ox) \tag{5}$$

$$\Delta E = E_s(1 - ox) - E_1 - s_c\ b\ (1 - d_f) \tag{6}$$

where E_s is the methane reaching the surface after S1 and d_f the destruction efficiency of the flare or engine. Because ΔE is linear in s_c, the cover reduces net emissions if and only if

$$s_c > s_c^* = -\frac{A_0}{A_1}, \quad A_0 = M_c(1 - ox) - (M_c - b)(1 - ox_{cov}) - b(1 - \rho\ ox), \quad A_1 = b(1 - \rho\ ox) - b(1 - d_f) \tag{7}$$

This is the quantitative form of the concern that sealing one area may only redirect leakage. Two structural alternatives are tested: redirected gas that is not oxidised ($\rho = 0$) and complete loss of oxidation under the cover ($\delta = 1$).

Scenario	Incremental change	Counterfactual
S0	Documented recovery, power generation and flaring	Reference
S1	Wellfield optimisation and leak detection	S0
S2	S1 + conventional engineered cover on hot-spot area	S1
S3	S1 + selective experimental coating	S1
S4	S2 + conventional photovoltaics on the covered area	S2
S5	S2 + experimental PV paint on the covered area	S4 at equal area
S6	S2 + gas buffer + biomethane upgrading (surplus + captured gas)	Flaring
S6b	S1 + gas buffer + biomethane upgrading, no cover	Flaring
S7	S6 + PV-assisted upgrader + adaptive control	S6
PW	Additional gas engines on the existing surplus (power alternative)	Flaring

Table 1. Scenario definitions. S1–S7 are the integrated interventions; PW is reported separately as the alternative use of the surplus gas.

3.4 Uses of recovered gas and their climate accounts

Three biomethane routes are priced: bio-GNC to an anchor fleet with a demand cap, pipeline injection with a spur and metering station, and a hybrid; the best route is chosen per draw. Electricity for upgrading (0.20 to 0.40 kWh per Nm³ raw gas) is common to all routes, and compression is route-specific: 0.25 kWh per Nm³ of product for GNC dispensing and 0.10 kWh per Nm³ for pipeline injection (Angelidaki et al., 2018). On the GNC-only route, product in excess of the anchor demand is flared, and its slip is counted. Plant fugitive methane is removed from the product stream before sale and counted as an emission (Bakkaloglu et al., 2022). Relative to flaring the same gas, the climate account of upgrading is

$$\Delta \mathrm{CO_2e} = EF_{NG}\ E_{\exp} - \mathrm{GWP}\ \Delta m_{\mathrm{CH_4}} - EF_{\mathrm{grid}}\ W_{\mathrm{el}} \tag{8}$$

with E_{exp} the energy of biomethane exported (displacing natural gas at 0.202 tCO_2 per MWh), Δm_{CH4} the change in methane emitted (buffer and plant leakage, tail-gas slip and flaring of unsold product, minus the flare slip avoided) and W_{el} the grid electricity used. The power alternative (PW) installs additional gas engines on the same surplus (Bove and Lunghi, 2006); its climate account is the grid electricity displaced. Gas already consumed by existing engines is never counted as feedstock. The first version of the model applied the GNC compression energy to every route, did not count unsold product or plant fugitive methane, and had no climate account for the power alternative; all four are corrected here.

3.5 Economics

Cash flows are in real US dollars over 20 years. The net present value is

$$\mathrm{NPV} = -\kappa \ \mathrm{CAPEX} + AF(r)\ (m_p R - O), \qquad AF(r) = \frac{1-(1+r)^{-20}}{r}, \qquad \kappa = 1 + f_{\mathrm{imp}}\ (d+l) \quad (9)$$

where κ is the capital premium built from the imported share fi_{mp}, the import duty d and logistics and exchange costs l, m_p a lognormal realised-price multiplier (mean 1, σ = 0.25) for tariff and exchange-rate risk, and r the real discount rate (8 to 18%, mode 12%, consistent with the 2025 country-risk spread). Replacement of short-lived assets (coatings, paint) is discounted explicitly. For photovoltaics on the covered area, setting the levelised cost of the paint equal to that of conventional PV gives the break-even efficiency

$$\eta^{*} = \frac{c_{pt}\ \kappa\ [CRF(r, L_{pt}) + o_{pt}]}{LCO\, E_{\mathrm{conv}}\ H_{poa}\ PR} \quad (10)$$

Carbon-credit revenue is zero in the base case. The levelised abatement cost is −NPV/(AF·ΔCO_2e) and equals the credit price at which a scenario breaks even. Variants (Section 4.8) add corporate tax at 35% with straight-line depreciation over 10 years (Argentina, 2021), a weighted average cost of capital built from debt and equity costs, contracted rather than merchant electricity, and expected credit revenue with an eligibility probability.

3.6 Monte Carlo design, balances and automated tests

Inputs (Appendix A) are sampled by Latin hypercube (McKay et al., 1979) with triangular, uniform or lognormal marginals and are tagged as operator-reported [R], literature or standard value [L], dated benchmark [B] or assumed [A]. Sensitivity is reported as Spearman rank correlation (Saltelli et al., 2008). For every scenario the model closes three balances: methane (generated = emitted + oxidised + destroyed + exported), carbon (the same flows in tonnes of carbon, with biogenic CO_2 from oxidation and combustion) and energy (offered = exported + flared + lost, with electricity imported tracked separately). The climate benefit of each scenario is computed both from the incremental formulas and from the balances, and the two must agree. 51 automated tests must pass before results are produced (Appendix B): unit conversions; closure of the cover, upgrader and scenario balances; agreement of the balance-based and incremental climate benefits; limiting cases (a barrier that blocks nothing avoids nothing; unchanged oxidation gives $s_c\, b(d_f - ox)$; zero gas prices make upgrading unprofitable); a hand-calculated upgrading case; an independent cash-flow benchmark of the annuity NPV; the analytic first-order-decay limits; LCOE parity at η^{*}; and a standard error of P(NPV > 0) below 0.01.

3.7 Robustness design

Beyond the three observation-error models, the analysis is repeated (120,000 draws each) with rank correlations induced by the Iman–Conover method (Iman and Conover, 1982) between gas and electricity prices,

between the prices and the costs that move with them, among capital costs, between the realised-price multiplier and logistics, and between L_0 and f_{ss}; with uniform instead of triangular marginals; with every prior widened by 50%; with the two structural cover alternatives; with contracted electricity; with a WACC; over a sweep of discount rates; and with expected credit revenue. Temporal variability is tested by replacing constant gas supply with a first-order-decay trajectory (IPCC, 2006),

$$\frac{G(t)}{G(0)} = \frac{1 - e^{-k(a+t)}}{1 - e^{-ka}} \quad (t < t_c), \qquad \frac{G(t)}{G(0)} = \frac{1 - e^{-k(a+t_c)}}{1 - e^{-ka}}\; e^{-k(t-t_c)} \quad (t \geq t_c) \tag{11}$$

with decay rate k (0.05 to 0.15 yr^{-1}), years of deposition a and years to closure t_c, and annual cash flows replacing the annuity for the uses of surplus gas.

4. Results

4.1 Calibration and the inferred collection efficiency

Reported flows imply that the existing engines consume 5,861 (4,931 to 6,878) Nm³ CH_4 h^{-1} of a collected 7,319 (6,390 to 8,291) (median, 90% interval), leaving a surplus of 1,452 (87 to 2,808) Nm³ CH_4 h^{-1}. The surplus is positive in 96% of draws and exceeds 500 Nm³ h^{-1} in 87%, but it depends strongly on engine capacity: the median is 2,232 Nm³ h^{-1} for installed power of 25 MW or less and 633 Nm³ h^{-1} once the expansion to about 33 MW is complete (Fig. 1d).

The prior gives generation around 2022 of 43,527 (24,087 to 72,266) Nm³ CH_4 h^{-1} and a modelled 2022 emission of 22 (10 to 39) t h^{-1}. The satellite constraint shifts generation to 49,642 (34,300 to 70,876) Nm³ h^{-1} and the emission at the observation date to 25 (16 to 38) t h^{-1}, and the three error models give nearly the same posterior (Fig. 1a,b): the inferred 2022 collection efficiency is 0.13 (0.09 to 0.19) (lognormal), 0.13 (0.10 to 0.18) (Gaussian) and 0.13 (0.09 to 0.17) (window). Carried forward with the 2025–26 capture, the efficiency is 0.14 (0.10 to 0.21) and the surface emission 26 (17 to 40) t h^{-1} (Fig. 1c). The collection efficiency of the first version, 0.18, compared a 2025–26 capture with 2019–22 satellite data and is superseded; it was never a measurement. One percentage point of collection efficiency corresponds to 3.3 (2.2 to 4.7) kt CH_4 yr^{-1} (about 88 kt CO_2e at GWP100), and the total baseline emission including slip from collected gas is 232 (148 to 349) kt CH_4 yr^{-1}, about 6.3 Mt CO_2e yr^{-1}.

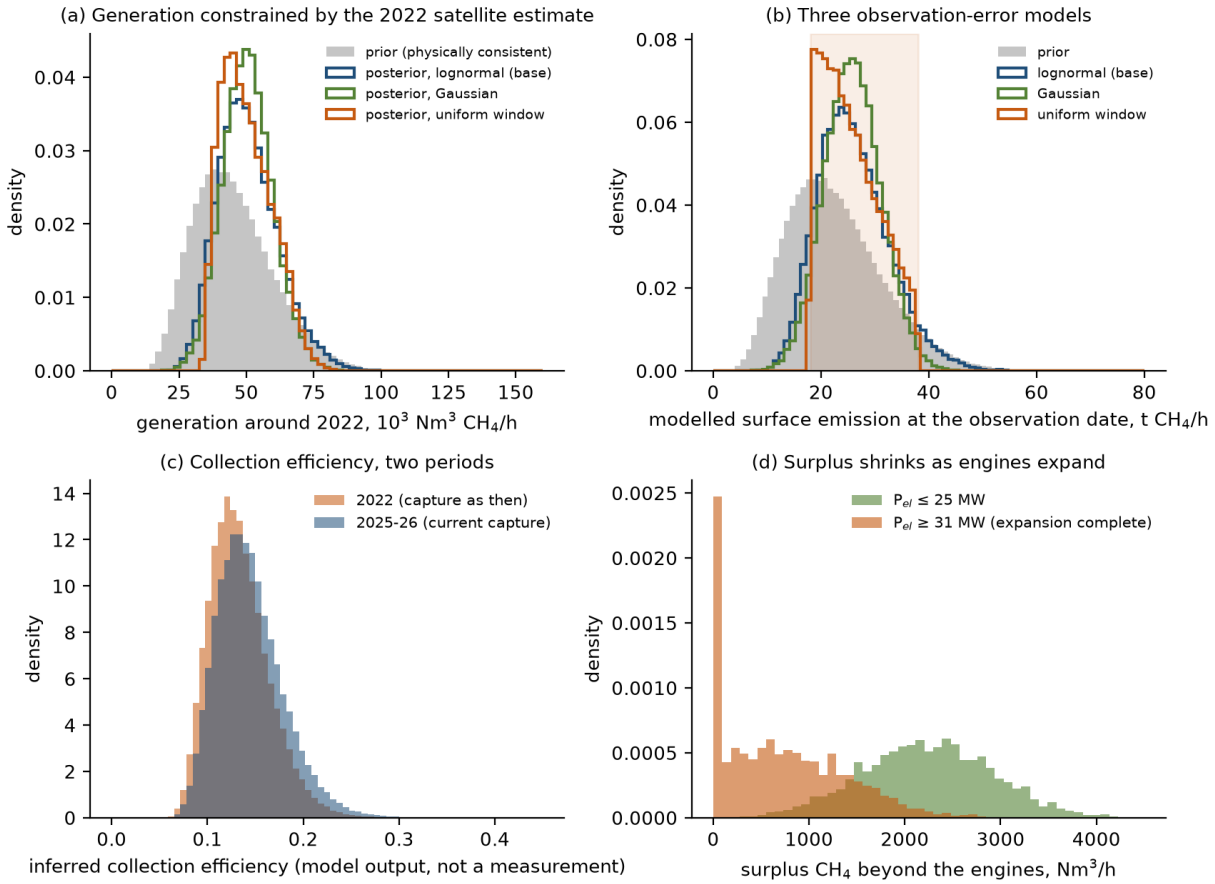


Fig. 1. Calibration. (a) Generation around 2022: prior and the posteriors of the three observation-error models. (b) Modelled surface emission at the observation date with the satellite range shaded. (c) Inferred collection efficiency in 2022 and 2025–26 (lognormal posterior; a model output, not a measurement). (d) Surplus methane beyond the existing engines by installed capacity.

4.2 Wellfield optimisation and selective covers

S1 avoids a median 176 (52 to 380) kt CO_2e yr^{-1} at a net present cost of 5.9 MUSD, a levelised abatement cost of US$4.6 per tCO_2e. A conventional cover reduces net methane emissions in 92% of draws and an experimental coating in 91% (Fig. 2a); the net effect of the cover alone is 8.1 (−1.1 to 25.3) kt CH_4 yr^{-1} for the conventional cover. Net avoided methane is 64% of the gross added capture (median), so gross capture overstates the benefit by about a third. The break-even captured share s_c^* has a median of 0.14 (0.04 to 0.30); the cover increases emissions in exactly the 7.5% of draws in which s_c falls below its own s_c^*, and the analytic boundary tracks the simulated sign change (Fig. 2b). The cover's effect is controlled by s_c (Spearman ρ = 0.74), then by fh_{ot}, ox and ε. Under the structural alternatives the probability of net benefit falls to 81% (redirected gas not oxidised) and 88% (all oxidation lost under the cover).

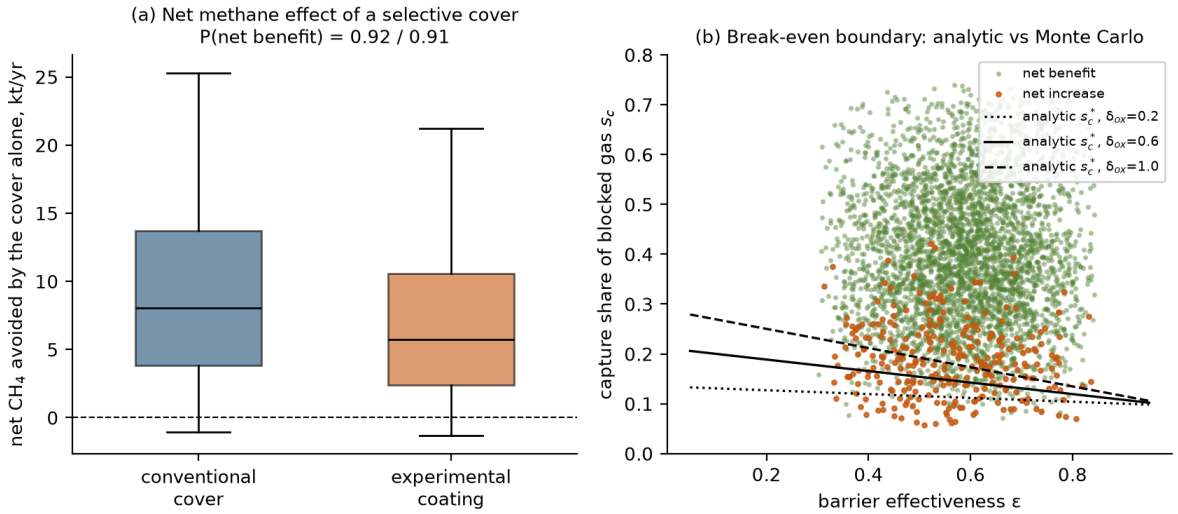


Fig. 2. Selective cover. (a) Net methane avoided by the cover alone (boxes P25–P75, whiskers P5–P95). (b) Simulated net benefit or increase against barrier effectiveness and captured share, with the analytic break-even s_c^* for three values of the oxidation-loss parameter.

4.3 Scenario economics without credits

None of the eight integrated intervention scenarios has a positive median incremental NPV under the assumed no-credit conditions (Fig. 3a; Table 3). The probability of a positive NPV is at most 0.014 (S6b). Measures do not add: S6, which upgrades the gas captured by the cover, has a lower median NPV than S6b, which upgrades without a cover (−19.4 against −14.7 MUSD), because the cover costs more than the extra gas is worth, and S7 avoids the most methane (512 kt CO_2e yr^{-1}) at the largest cost. Corporate tax with depreciation shields reduces the losses but does not change any sign (Table 3, after-tax column).

The power alternative is reported separately because it is not an emission-reduction intervention but a competing use of gas that is already collected. Additional engines on the existing surplus have a positive NPV in 98.5% of the draws with surplus gas at merchant tariffs (median 6.3 (0.7 to 16.4) MUSD) and in 100.0% with a RenovAr-type contract (median 14.9 (3.5 to 30.9) MUSD; Fig. 3b). Their climate benefit, from displaced grid electricity, is 8.9 (0.6 to 10.8) kt CO_2e yr^{-1}, an order of magnitude smaller than that of the capture measures.

Scenario	NPV	P(NPV>0)	After tax	Avoided (kt CO_2e/yr)	Abatement	P(>0), US$25
S1	−5.9 (−8.9 to −3.6)	0.000	−4.2	176 (52 to 380)	4.6	0.98
S2	−10.6 (−17.6 to −6.6)	0.000	−7.8	409 (109 to 931)	3.6	0.97
S3	−17.7 (−39.0 to −9.4)	0.000	−13.0	348 (93 to 820)	7.2	0.93
S4	−17.5 (−36.3 to −8.9)	0.001	−14.7	415 (115 to 939)	6.0	0.95
S5	−23.0 (−69.0 to −8.1)	0.002	−17.7	413 (113 to 937)	8.0	0.89
S6	−19.4 (−29.1 to −8.7)	0.005	−15.3	418 (119 to 941)	6.2	0.95
S6b	−14.7 (−22.2 to −4.5)	0.014	−11.6	186 (60 to 389)	10.5	0.89
S7	−21.5 (−31.1 to −10.7)	0.002	−17.0	512 (192 to 1,061)	5.6	0.98
PW	6.1 (0.0 to 16.3)	0.946	3.1	9 (1 to 11)	(profit)	0.96

Table 3. Scenario results, lognormal posterior (median, P5 to P95). NPV incremental to S0 in MUSD with no credit revenue; after-tax median with 35% tax and 10-year depreciation; avoided emissions at GWP100; abatement cost in US$ per tCO₂e; last column P(NPV > 0) with a US$25 credit. PW: additional engines on the existing surplus.

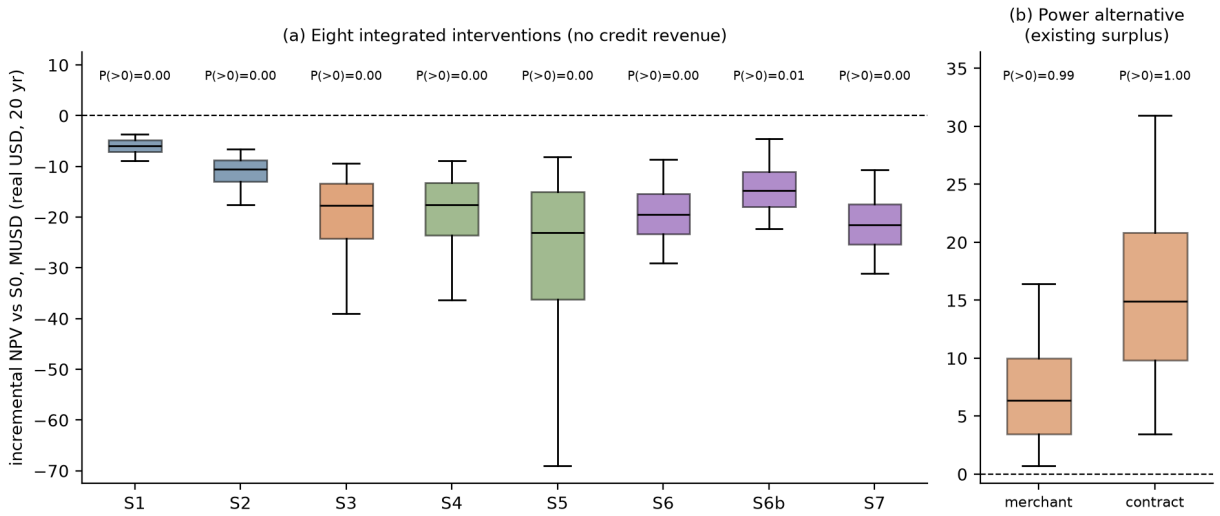


Fig. 3. (a) Incremental NPV of the eight integrated interventions, lognormal posterior, no credit revenue (boxes P25–P75, whiskers P5–P95). (b) NPV of additional engines on the existing surplus at merchant and contracted tariffs.

4.4 Biomethane against additional power

For the surplus beyond the existing engines, biomethane has a positive NPV in 5.4% of draws (median −9.8 (−16.3 to 0.3) MUSD) and beats additional engines in 0.02%. Where it is built, bio-GNC to an anchor customer is the preferred route in 99.9% of draws, because the pipeline price anchored to the PIST benchmark is low relative to the cost of a spur. The break-even realised bio-GNC plant-gate price is US$0.46 (0.26 to 1.42) per Nm³ (about US$14 per MMBtu) against a prior of 0.13 to 0.40; pipeline injection would need US$0.60 per Nm³ (US$18 per MMBtu), about six times the PIST price. To match additional engines, bio-GNC would need US$0.61 per Nm³. Sizing does not rescue upgrading: with only the existing surplus as feedstock, the probability of a positive NPV peaks at 0.05 at 2,000 Nm³ h^{-1} of raw gas (Fig. 4b). The climate accounts of the two uses are similar: upgrading avoids more CO_2e than power in 54% of draws, by a median of only 199 t yr^{-1}. A gas buffer lowers NPV in 99.97% of draws (median −1.18 MUSD).

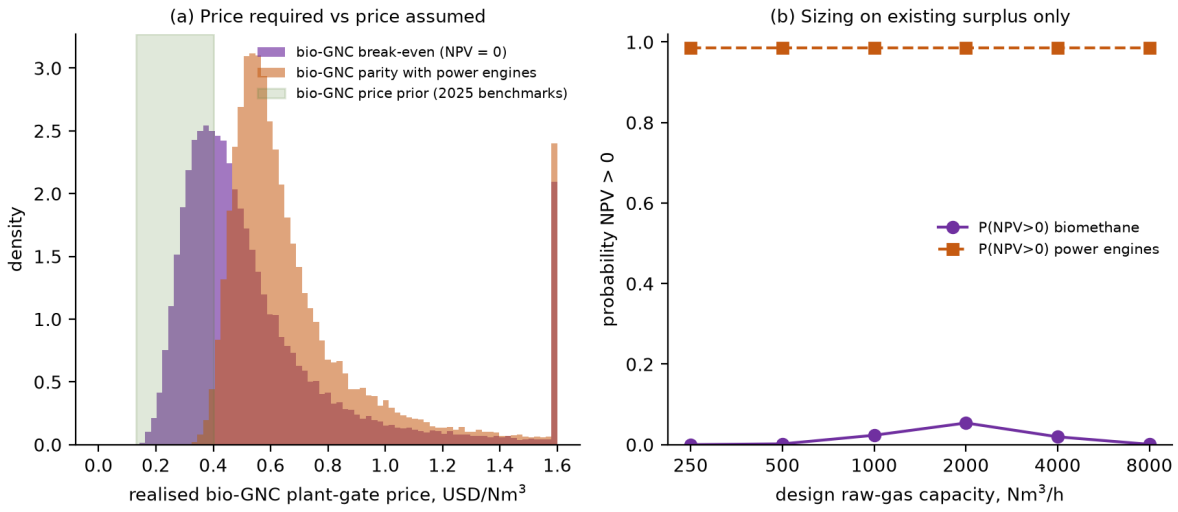


Fig. 4. (a) Bio-GNC plant-gate price required for NPV = 0 and for parity with additional engines, against the prior (values above US$1.6 per Nm³ pooled). (b) Probability of positive NPV against design capacity, existing surplus only.

4.5 Photovoltaic coating against conventional photovoltaics

On the covered area, conventional PV has a positive NPV in 4.0% of draws and a median LCOE of US$119 (84 to 169) per MWh, above the merchant value of electricity because of the landfill foundation premium, the capital premium and the discount rate. The paint, with deliberately wide priors (efficiency 2 to 10%, installed cost US$5 to 150 m^{-2}, life 2 to 20 years), has a median LCOE of US$211 (60 to 605) per MWh and is cheaper in 23% of draws. The break-even efficiency η^*, Eq. (10), has a median of 9.7% (2.9% to 24.9%) and exceeds 10% in 48% of draws (Fig. 5a). The 23% figure is a property of unvalidated priors; the transferable result is the η^* surface.

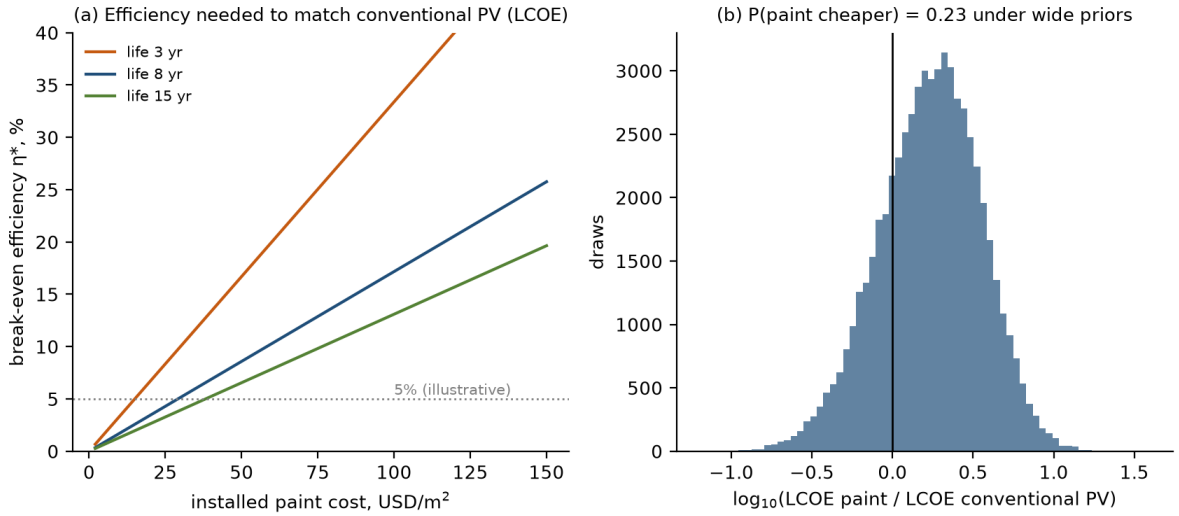


Fig. 5. (a) Break-even efficiency of PV paint for LCOE parity with conventional PV against installed cost, for three lifetimes (other inputs at posterior medians). (b) Distribution of the LCOE ratio.

4.6 Abatement cost

Median levelised abatement costs are low in absolute terms: US$3.6 (S2), 4.6 (S1), 5.6 (S7) and 10.5 (S6b) per tCO_2e at GWP100 (Fig. 6). At GWP20 the avoided CO_2e is 2.95 times larger and the cost correspondingly lower; the median avoided in S2 becomes 1,206 kt CO_2e yr^{-1}.

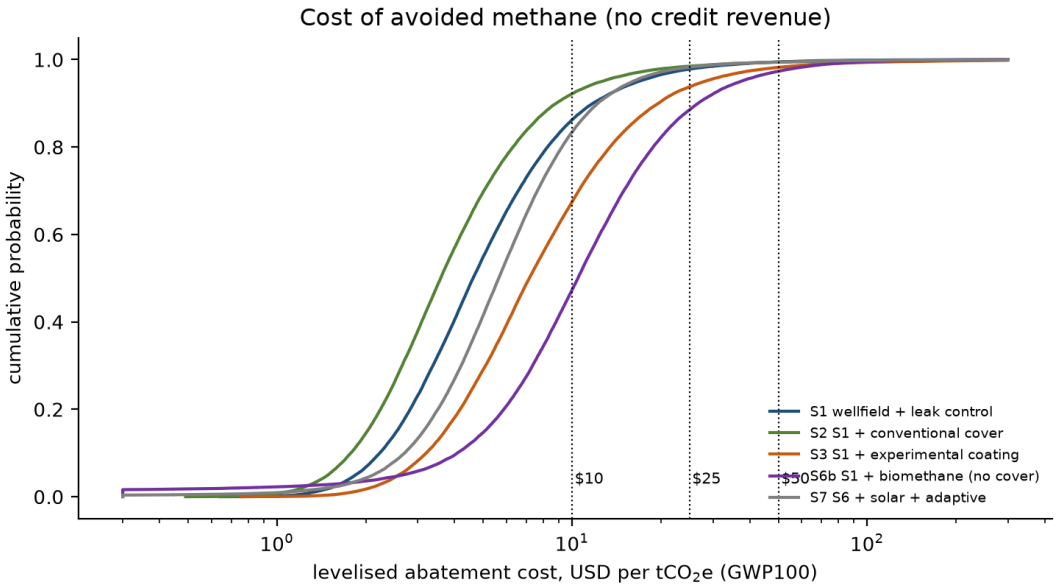


Fig. 6. Cumulative distribution of levelised abatement cost (GWP100, no credit revenue); dotted lines mark credit prices of US$10, 25 and 50 per tCO₂e.

4.7 Sensitivity

The NPV of upgrading without a cover (S6b) is controlled by price and capital: the realised-price multiplier (ρ = 0.52), the bio-GNC price (0.52) and the upgrader capital cost (−0.38); gas-side inputs rank lower (Fig. 7a). The value of additional engines depends on the same price multiplier (0.64), then on the methane fraction, the discount rate and the installed engine capacity (Fig. 7b). Tariff and exchange-rate risk, not gas chemistry, is the dominant uncertainty for monetising recovered gas.

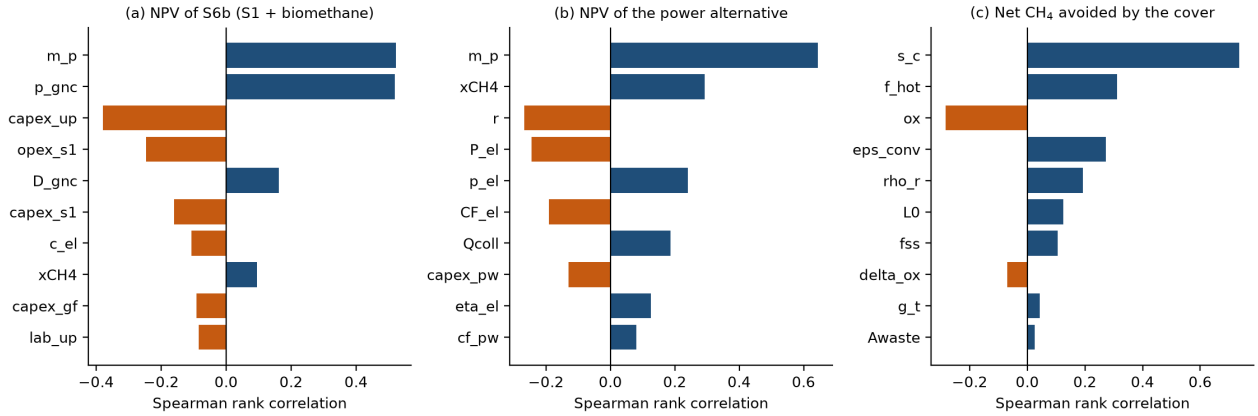


Fig. 7. Spearman rank correlations, lognormal posterior: (a) NPV of S6b; (b) NPV of the power alternative; (c) net methane avoided by the conventional cover. Top ten inputs.

4.8 Credits, tax, financing and the discount rate

Because the abatement costs are low, a credit near US$5 to 10 per tCO₂e would make the capture measures viable if the added capture were eligible: at US$25 the probability of a positive NPV is 0.98 for S1 and 0.97 for S2 (Fig. 8a). At the 2025 market range, with an eligibility probability between 0.2 and 0.8, the expected credit

revenue raises P(NPV > 0) only to 0.33 (S1) and 0.38 (S2). The discount rate does not change any conclusion between 6 and 20% real (Fig. 8b): S1 and S2 remain negative at every rate, S6b reaches at most 0.049, and the power alternative stays above 0.90. A WACC built from equity and debt costs gives a median S6b NPV of −14.6 MUSD against −14.7 in the base case. CEAMSE reports that it began certifying carbon credits linked to biogas management in 2025 (Canal 26, 2026); whether the added capture modelled here would be additional, verifiable and saleable is outside this study.

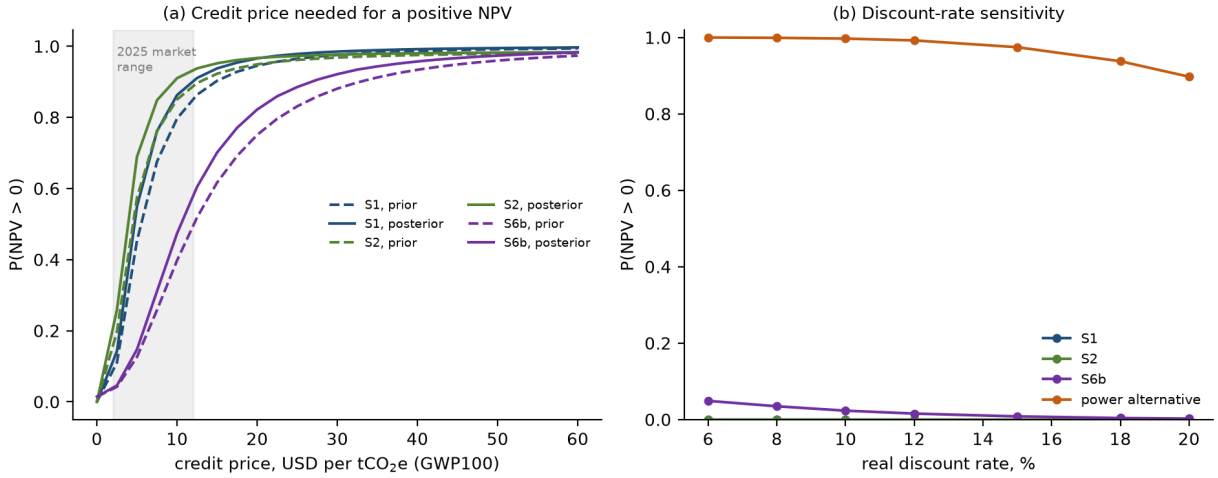


Fig. 8. (a) Probability of positive NPV against credit price for S1, S2 and S6b, prior (dashed) and lognormal posterior (solid), with the 2025 market range shaded. (b) Probability of positive NPV against the real discount rate.

4.9 Robustness

Table 4 and Fig. 9 repeat the headline quantities under every variant. The decision-relevant conclusions do not change: no integrated scenario has a positive median NPV in any variant, additional engines on the surplus remain profitable in 92% to 100% of draws, upgrading beats them in under 1% of draws, and the cover's net benefit stays between 81% and 92%. The climate case moves with the generation prior and the emission estimate: without the satellite constraint S2 avoids 338 kt CO_2e yr^{-1} instead of 405, and widening every prior by 50% raises it to 484. Correlated inputs change no headline quantity by more than a few per cent. With a first-order-decay supply trajectory, the median NPV of additional engines falls from 6.5 to 4.2 MUSD and remains positive in 93% of draws, while upgrading falls from −9.7 to −10.7 MUSD.

Variant	η 2022	η 2025–26	S2 avoided	P(cover)	S6b NPV	P(PW>0)	p*	Abatement S2	Any > 0
likelihood: lognormal	0.130	0.140	405	0.92	−14.7	0.99	0.46	3.7	no
likelihood: Gaussian	0.129	0.139	410	0.92	−14.7	0.99	0.46	3.6	no
likelihood: uniform window	0.132	0.142	407	0.92	−14.7	0.99	0.46	3.6	no
prior only (no satellite constraint)	0.149	0.160	338	0.92	−14.7	0.99	0.46	4.4	no
correlated inputs (Iman-Conover)	0.134	0.144	402	0.92	−14.7	0.98	0.46	3.7	no
uniform marginals	0.120	0.130	439	0.83	−15.9	0.97	0.52	3.9	no
priors widened by 50%	0.122	0.132	484	0.90	−16.4	0.92	0.51	3.4	no
cover: redirected gas unoxidised	0.130	0.140	340	0.81	−14.7	0.99	0.46	4.4	no
cover: all oxidation lost under cover	0.130	0.140	373	0.88	−14.7	0.99	0.46	4.0	no
contracted electricity tariff (RenovAr-type)	0.130	0.140	405	0.92	−14.7	1.00	0.46	3.7	no
WACC from equity, debt and tax	0.130	0.140	405	0.92	−14.6	0.98	0.47	3.7	no

Table 4. Robustness of the headline quantities (medians unless stated). η: inferred collection efficiency; S2 avoided in kt CO_2e yr^{-1}; P(cover): probability that the cover reduces net emissions; S6b NPV in MUSD; P(PW > 0): probability that additional engines have positive NPV; p: bio-GNC break-even price in US$ per Nm^3; abatement cost of S2 in US$ per tCO_2e; last column: does any integrated scenario have a positive median NPV?*

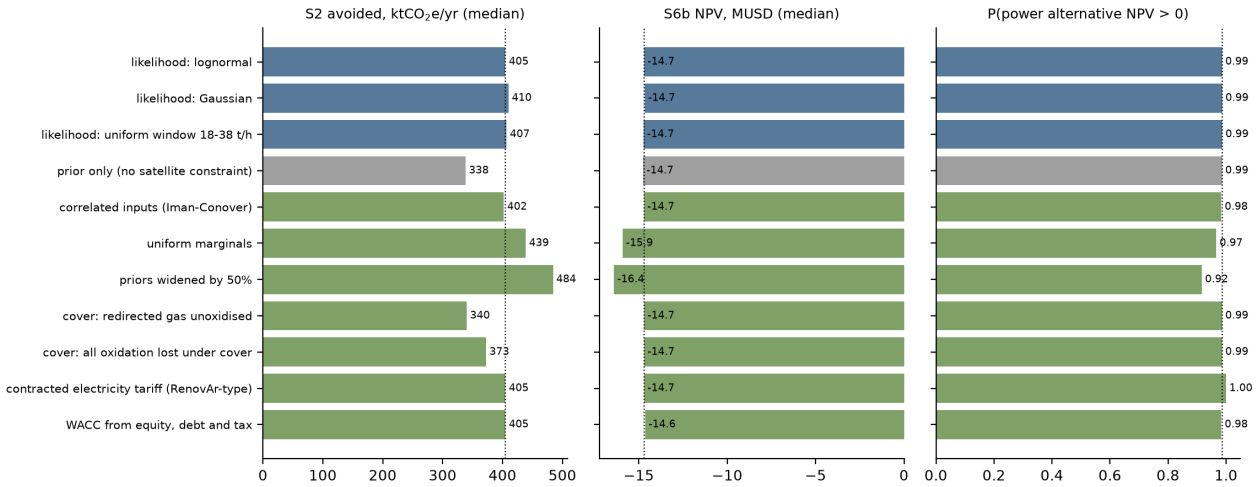


Fig. 9. Headline quantities under the observation-error models (blue), without the satellite constraint (grey) and under the other robustness variants (green); the dotted line marks the base case.

4.10 Physical consistency: methane, carbon and energy balances

In every scenario the methane, carbon and energy balances close to round-off (relative residuals below 10^{-9}), and the climate benefit computed from the balances equals the incremental benefit to within 10^{-8} (Appendix B). Fig. 10a shows the fate of the generated methane for a median draw: in the reference case 232 kt CH_4 yr^{-1} is emitted, 48 kt is oxidised in the cover soil and 45 kt is destroyed in engines and flares. The measures shift methane from the first category to the third; upgrading exports 5.0 kt CH_4 yr^{-1} as biomethane and adds a fossil-CO_2 credit (displaced natural gas minus grid electricity used) that is small against the methane term (Fig. 10b).

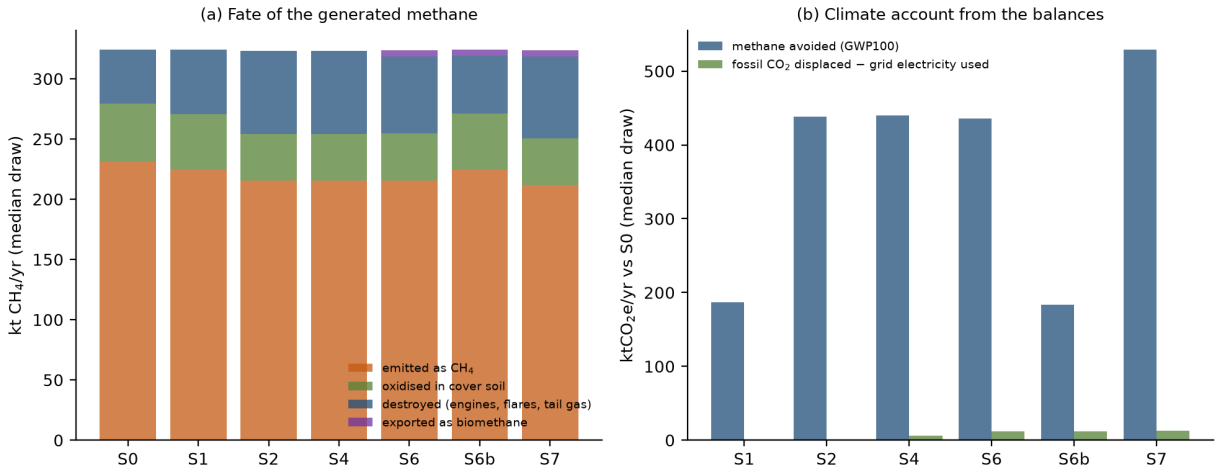


Fig. 10. (a) Fate of the generated methane by scenario (median draw). (b) Climate account from the balances: methane avoided (GWP100) and fossil CO_2 displaced net of grid electricity used, relative to S0.

4.11 Scorecard for the research hypotheses

Hypothesis	Result
H1: cover plus pressure-managed extraction lowers total emissions	Supported conditionally: net reduction in 92% of draws if $s_c > s_c^*$ (median 0.14); net benefit is 64% of gross capture.
H2: PV coatings beat conventional PV only above measurable thresholds	Paint cheaper in 23% of draws under wide, unvalidated priors; break-even efficiency 9.7% at median inputs.
H3: biomethane from surplus gas has higher risk-adjusted NPV than alternatives	Not supported: beats additional engines in 0.02% of draws; break-even US$0.46 per Nm^3 (US$14 per MMBtu).
H4: buffering offsets its leakage, energy and capital costs	Not supported: positive in 0.03% of draws; median ΔNPV −1.18 MUSD.
H5: the integrated configuration has lower lifecycle CO_2e per tonne of waste	Supported but small: S7 lower in 99.6% of draws; median reduction 8.3%, from 1.04 to 0.95 tCO_2e per tonne.

Table 5. Outcome of the five hypotheses of the project white paper (lognormal posterior).

5. Discussion

5.1 What the model says about priorities

The largest finding concerns scale. Collected methane is about 5.2 t h^{-1}, whereas the satellite-constrained generation is about 37 t h^{-1}, so most generated methane is not collected. The measures that change the balance are those that raise collection efficiency, at about 3.3 kt CH_4 yr^{-1} per percentage point; coatings, PV and buffers act on the margin. The integrated package S7 lowers per-tonne emissions by a median 8%. This conclusion follows from the reported capture and the satellite constraint and does not depend on the economic priors. Because measured collection efficiencies at active sites span a wide range (Spokas et al., 2006; Giordano et al., 2024; Saul et al., 2026), a site flux campaign is the measurement with the largest value of information.

5.2 Decisions that are robust, and decisions that are not

Robust under the stated priors and every variant: no integrated measure pays without credit revenue; abatement is cheap per tonne; a cover needs a minimum captured share to help; a buffer does not pay; additional engines on genuinely uncommitted surplus gas are profitable at merchant and contracted tariffs. Not robust: the existence of surplus gas once the engine expansion is complete; the ranking of the capture measures by abatement cost, which depends on the emission estimate; and the comparison of paint with PV, which depends on three unvalidated numbers. The break-even prices, s_c^* and η^* are the portable outputs: a reader with a dated quotation or a measured captured share can read the decision from them. We recommend that no capital be committed to coatings, buffers or upgrading before three measurements are in hand: a calibrated site flux survey to fix the collection efficiency, hourly gas flow and engine dispatch records to fix the true surplus, and a firm offtake or tariff quotation.

5.3 Limitations

The model is a lumped-parameter screening tool. It has no spatial gas-flow solution, so the hot-spot fraction, barrier effectiveness and captured share are parameters, not predictions; a pressure-resolved porous-media model is the next step for the cover question. The satellite constraint rests on a few retrievals from 2019–22 with wind and retrieval uncertainty, and the three error models span but do not exhaust the possibilities; repeated observations of the site after the capture increase would sharpen the inference most. The generation prior is an assumption, as is the growth factor that carries it to 2025–26. Operator figures were taken as reported, and the 23 MW and 33 MW capacities come from the white paper's summary of the operator statement. Prices are anchored to the benchmarks of Table 2, but peso conversions and the plant-gate price a bio-GNC buyer would actually pay

are assumptions. The PV-paint concept derives from the author's unpublished competition submission, and no experimental data exist for it; treating its performance as a prior asks what it would need to achieve, not what it will achieve.

5.4 What would falsify these conclusions

Conclusion	Fails if
Covers reduce net methane	The captured share of blocked gas falls below s_c^* (median 0.14), for example when extraction headroom or header capacity is limiting.
Biomethane loses to power	A firm bio-GNC price above about US$0.61 per Nm³, or electricity worth less than the engines' break-even (median US$39 per MWh), or no engine capacity can be added.
Surplus gas exists	The 33 MW expansion is commissioned (surplus positive in 80% of those draws, median 633 Nm³ CH_4 h^{-1}).
Collection efficiency near 0.14	A calibrated flux survey after the 2025–26 capture increase finds a surface emission outside about 17 to 40 t h^{-1}.
PV paint is unlikely to beat PV	Validated efficiency exceeds η^* for its measured cost and life (Fig. 5a).

Table 6. Conditions under which each conclusion fails.

6. Conclusions

1. A methane balance constrained by satellite estimates at their own observation date implies a collection efficiency of about 0.13 in 2022 and 0.14 in 2025–26 at Norte III; these are inferences conditional on a wide generation prior and agree across three observation-error models. The leading mitigation lever is collection efficiency, worth about 3.3 kt CH_4 yr^{-1} per percentage point.

2. None of the eight integrated intervention scenarios has a positive median incremental NPV under the assumed no-credit conditions, although the median abatement costs are US$3.6 to 10.5 per tCO_2e; credit eligibility, not engineering feasibility, decides bankability.

3. Additional engines on uncommitted surplus gas, the competing use reported separately, are profitable in 99% of draws at merchant tariffs; biomethane would need about US$0.46 per Nm³ (US$14 per MMBtu) to break even.

4. A selective cover helps only when the captured share of the gas it blocks exceeds a closed-form threshold (median 0.14); gross capture overstates the net benefit by about a third.

5. Every scenario closes its methane, carbon and energy balances, and the conclusions survive correlated inputs, alternative priors, structural cover alternatives, dated tax and financing assumptions and time-varying gas supply.

Declarations

CRediT authorship contribution statement. Leon Sandler: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review and editing.

Declaration of generative AI and AI-assisted technologies in the writing process. During the preparation of this work the author used Claude (Anthropic) to assist with source gathering, code development, drafting and editing. The author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Declaration of competing interest. The author declares no known competing financial interests or personal relationships that could have influenced the work reported in this paper. The PV-coating concept examined in Section 4.5 originates from the author's own unpublished competition submission.

Data and code availability. The model (NorteIII_MC.py), the automated tests, the figure and export code (NorteIII_figs.py), the parameter table and all results are available at https://github.com/sandlerleon/norte3-landfill-methane-decision and archived at https://doi.org/10.5281/zenodo.23823891 (software, version 2.0.0); this manuscript is archived at https://doi.org/10.5281/zenodo.23823886. One command regenerates every number, table and figure.

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Appendix A. Input distributions

Table A1 lists the sampled inputs. Distributions are triangular (minimum, mode, maximum), uniform (minimum, maximum) or lognormal (σ, mean 1). Tags: R, operator-reported or published programme value; L, literature or standard value; B, dated market or regulatory benchmark (Table 2); A, assumed by the author and requiring a site-specific or dated quotation before use in a decision. Inputs marked * enter only the robustness variants.

Input	Description	Distribution	Tag
Qcoll22	collected biogas around 2022 (satellite period), Nm3/h; 13,100 peak in 2013, 13,800 in 2025	tri(12,000, 13,100, 14,000)	A
Awaste	waste to Norte III, t/day (~18,000 reported)	tri(14,000, 16,500, 19,000)	R
L0	CH4 generation potential, Nm3 CH4 per t waste (inventory 100, CAA 170)	tri(50, 100, 170)	L
fss	fraction of steady-state generation reached (age/decay/lag)	tri(0.3, 0.6, 0.95)	A
xCH4	CH4 fraction of landfill gas	tri(0.42, 0.5, 0.57)	A
ox	cover methane oxidation fraction	tri(0.05, 0.15, 0.35)	L
g_t	generation 2025-26 relative to 2022	tri(0.95, 1.05, 1.15)	A
Qcoll	collected biogas 2025-26, Nm3/h (13,800 in 2025; 16,000 Apr 2026)	tri(13,000, 14,800, 16,500)	R
P_el	installed biogas power, MW (23 in 2020; 33 planned 2026)	tri(23, 28, 33)	R
CF_el	engine capacity factor	tri(0.7, 0.85, 0.93)	A
eta_el	engine electrical efficiency	tri(0.36, 0.4, 0.43)	L
d_f	flare/engine CH4 destruction efficiency	tri(0.95, 0.98, 0.995)	L
dEta1	S1 absolute gain in collection efficiency (of generation)	tri(0, 0.02, 0.06)	A
capex_s1	S1 capex, MUSD	tri(0.5, 1.5, 4)	A
opex_s1	S1 opex, MUSD/yr	tri(0.15, 0.4, 1)	A
f_hot	flux fraction passing covered area	tri(0.1, 0.25, 0.5)	A
h_e	hot-spot flux enrichment over mean	tri(1.5, 3, 6)	A
A_act	actively emitting area, ha	tri(100, 200, 400)	A
eps_conv	conventional cover: fraction of arriving CH4 blocked	tri(0.3, 0.6, 0.85)	A
eps_coat	experimental coating: fraction blocked	tri(0.1, 0.45, 0.85)	A
s_c	share of blocked CH4 captured by extraction	tri(0.05, 0.35, 0.75)	A
delta_ox	fractional loss of oxidation under cover	tri(0.2, 0.6, 1)	A
rho_r	oxidation of redirected gas relative to ox	uni(0, 1)	A
c_conv	conventional cover cost, kUSD/ha	tri(80, 150, 300)	A
c_coat	experimental coating cost, kUSD/ha	tri(100, 250, 700)	A
om_cov	cover O&M, fraction of capex per yr	tri(0.015, 0.03, 0.05)	A
life_conv	conventional cover life, yr	tri(15, 25, 30)	A
life_coat	coating life, yr	tri(4, 8, 15)	A
r	real USD discount rate (EMBI 2025: 561 to >1,200 bps; 569 bps Dec 2025)	tri(0.08, 0.12, 0.18)	B
f_imp	imported share of equipment capex	tri(0.5, 0.7, 0.9)	A
duty	import duty on capital goods (0% BK regime; 12.6% Decree 513/2025; 20% Mercosur CET)	tri(0, 0.126, 0.2)	B
logist	freight, logistics and FX cost on imports, fraction	tri(0.03, 0.08, 0.15)	A
m_p	realised-price multiplier (tariff/FX risk), mean 1	logn(0.25)	A
avail_firm	fraction of surplus gas contractually firm	tri(0.6, 0.85, 1)	A
uptime	plant availability	tri(0.8, 0.9, 0.95)	A
rec	CH4 recovery in upgrading	tri(0.9, 0.95, 0.985)	L
tail_ox	fraction of tail-gas CH4 slip oxidised	uni(0, 0.95)	A
fug_plant	plant fugitive CH4, fraction of CH4 in	tri(0.002, 0.006, 0.012)	L
capex_up	upgrader capex, USD per Nm3/h raw capacity	tri(1,800, 3,000, 5,000)	A
e_up	upgrading electricity, kWh/Nm3 raw	tri(0.2, 0.28, 0.4)	L
c_el	electricity purchase cost, USD/MWh (wholesale monomic ~87-91 USD/MWh 2024-25	tri(70, 95, 130)	B

Input	Description	Distribution	Tag
	plus network)		
om_up	upgrader O&M, fraction of capex per yr	tri(0.03, 0.04, 0.06)	A
lab_up	labour/other, MUSD/yr	tri(0.15, 0.25, 0.45)	A
capex_gv	GNC compression capex, USD per Nm3/h product	tri(400, 800, 1,500)	A
capex_gf	GNC fixed capex, MUSD	tri(0.3, 0.8, 2)	A
p_gnc	bio-GNC net plant-gate price, USD/Nm3 (delivered station gas ~0.15, retail ~0.40-0.45, 2025)	tri(0.13, 0.19, 0.4)	B
p_pipe	pipeline net plant-gate price, USD/Nm3 (PIST 2.94-2.98 USD/MMBtu, Dec 2025)	tri(0.07, 0.1, 0.16)	B
D_gnc	bio-GNC anchor demand, Nm3/h	tri(300, 1,000, 3,000)	A
dist_km	distance to injection point, km	tri(2, 8, 20)	A
c_km	pipeline spur cost, MUSD/km	tri(0.3, 0.6, 1.2)	A
c_inj	injection/metering station, MUSD	tri(0.5, 1, 2)	A
capex_pw	gas-engine power capex, USD/kW (RenovAr landfill-gas ref 1.3 MUSD/MW)	tri(1,100, 1,300, 1,700)	R
p_el	electricity sale price, merchant/MAT, USD/MWh (marginal cost 69, monomic ~87-91, 2024-25)	tri(60, 75, 95)	B
p_el_k*	electricity sale price, RenovAr-type contract, USD/MWh (landfill-gas awards 128-130, 2017)	tri(110, 125, 135)	B
cf_pw	new engines capacity factor	tri(0.8, 0.88, 0.93)	A
eta_pw	new engines electrical efficiency	tri(0.36, 0.4, 0.43)	L
om_pw	engine O&M, fraction of capex per yr	tri(0.03, 0.045, 0.07)	A
var_loss	flow-variability loss without buffer	uni(0.05, 0.25)	A
zeta	fraction of variability loss removed by buffer	uni(0.5, 0.9)	A
capex_buf	buffer capex, MUSD	tri(0.3, 0.8, 2)	A
leak_buf	buffer CH4 leakage, fraction of throughput	tri(0.002, 0.005, 0.01)	A
f_pv	share of upgrader electricity from on-site PV	uni(0.1, 0.5)	A
dEta_ad	adaptive-control gain in collection efficiency	tri(0, 0.01, 0.03)	A
capex_ad	adaptive control capex, MUSD	tri(0.2, 0.5, 1.2)	A
opex_ad	adaptive control opex, MUSD/yr	tri(0.05, 0.1, 0.25)	A
kwp_ha	conventional PV, kWp per ha	tri(500, 700, 900)	A
capex_pv	PV capex, USD/kW (RenovAr ref 850)	tri(550, 750, 1,100)	R
prem_pv	landfill foundation premium for PV	tri(1.1, 1.3, 1.6)	A
yield_pv	PV yield, kWh/kWp/yr	tri(1,350, 1,500, 1,650)	A
om_pv	PV O&M, fraction of capex per yr	tri(0.01, 0.015, 0.025)	A
life_pv	PV life, yr	tri(20, 25, 30)	A
p_pv	PV electricity value, merchant, USD/MWh (2024-25 wholesale cost)	tri(55, 70, 90)	B
eta_pt	PV-paint module efficiency (unvalidated)	tri(0.02, 0.05, 0.1)	A
c_pt	PV-paint installed cost, USD/m2 (unvalidated; wide on purpose)	tri(5, 30, 150)	A
life_pt	PV-paint life, yr (unvalidated; wide on purpose)	tri(2, 6, 20)	A
pr_pt	PV-paint performance ratio	tri(0.65, 0.78, 0.85)	A
om_pt	PV-paint O&M, fraction of capex per yr	tri(0.01, 0.02, 0.04)	A
H_poa	plane-of-array irradiation, kWh/m2/yr	tri(1,550, 1,650, 1,750)	A
ef_grid	grid emission factor, tCO2/MWh	tri(0.28, 0.31, 0.36)	A
cp*	carbon-credit price if eligible, USD/tCO2e (VCM ~5.5, landfill gas ~9.5 USD, 2025)	tri(2, 6, 12)	B
p_elig*	probability that added capture is eligible for credits	uni(0.2, 0.8)	A
dv*	debt share of financing	tri(0, 0.4, 0.6)	A
r_e*	real cost of equity	tri(0.12, 0.16, 0.22)	A
r_d*	real cost of debt	tri(0.07, 0.09, 0.12)	A
k_dec*	first-order decay rate, 1/yr (IPCC 2006, temperate wet MSW)	tri(0.05, 0.09, 0.15)	L
age*	years of deposition before 2026	tri(20, 28, 32)	A
t_close*	years until deposition stops, from 2026	tri(3, 8, 15)	A

Table A1. Input distributions.

Appendix B. Balances and automated tests

Scenario	Emitted	Oxidised	Destroyed	Exported	Biogenic C	Exported energy	Electricity	Total CO_2e
S0	231.5	47.7	44.8	0.0	69.2	0.0	0.0	6.25
S1	224.6	46.2	53.2	0.0	74.5	0.0	0.0	6.06
S2	215.3	39.1	68.9	0.0	82.0	0.0	0.0	5.81
S4	215.3	39.1	68.9	0.0	82.0	0.0	0.0	5.81
S6	215.4	39.1	64.0	5.0	78.3	69.4	6.3	5.80
S6b	224.7	46.2	48.2	5.0	70.9	69.2	6.2	6.06
S7	211.9	38.3	68.3	5.0	80.9	69.4	4.3	5.71

Table B1. Scenario balances for the median draw: methane emitted, oxidised, destroyed and exported (kt CH_4 yr^{-1}); biogenic carbon released as CO_2 (kt C yr^{-1}); biomethane energy exported and grid electricity used (GWh yr^{-1}); total emissions (Mt CO_2e yr^{-1}, GWP100). Residuals of all balances are below 10^{-9} of generation.

The automated test suite (python NorteIII_MC.py --test) contains 51 checks, all of which pass: unit conversions (methane mass, carbon content, annuity and capital recovery factors); Latin-hypercube moments and the induced rank correlation; the white paper's screening arithmetic; closure of the cover balance; the analytic break-even s_c^*; two limiting cases of the cover; methane and carbon closure of seven scenario balances; methane and energy closure of four upgrader accounts; equality of balance-based and incremental climate benefits for four scenarios; a hand-calculated upgrading case; an explicit cash-flow benchmark of the annuity NPV; unprofitability at zero gas prices; LCOE parity at η^*; the calibration window; physical consistency of the posterior; determinism; the steady-state limit of the time-resolved NPV and the analytic slow-decay limit of the first-order-decay model; and the Monte Carlo standard error.

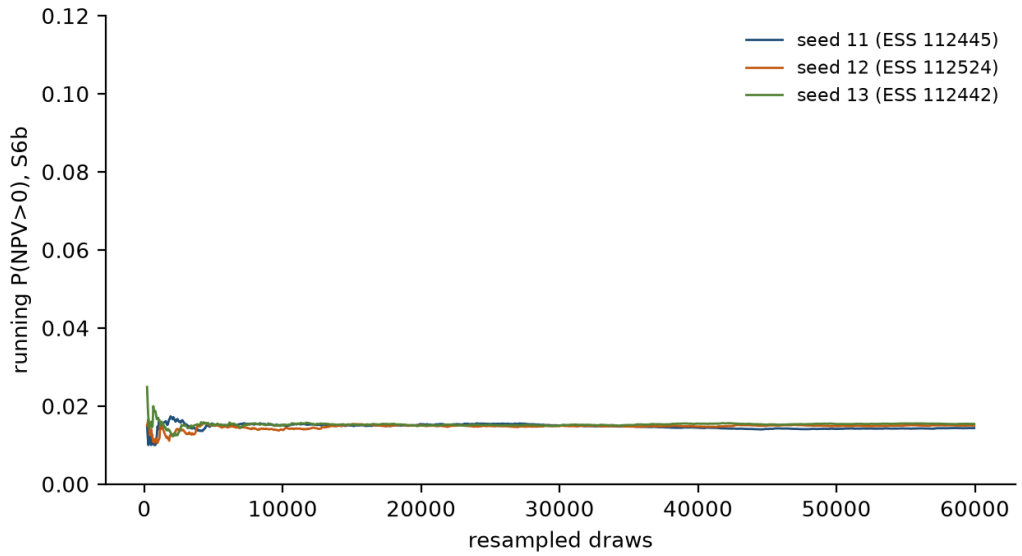


Fig. B1. Running probability of positive NPV for S6b across three independent seeds (resampled lognormal posterior; ESS in the legend).